METRIC
DOD-C-63537A(AR)
18 July 1984
SUPERSEDING
DOD-C-63537(AR)
25 January 1982

MILITARY SPECIFICATION

* CARTRIDGE, CALIBER .22, LONG RIFLE, TRACER, M861

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, and the assembly, and packing of Cartridge, Caliber .22, Long Rifle, Tracer, M861.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specifications For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109	-	Quality Assurance Terms and Definitions
* MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1168	-	Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

* 7553769	-	Packing and Marking - Cartridges, Cal .22, Long Rifle: Cartons and Cartons in Containers, Box, Ammunition, M2A1; Box, Wirebound
7643674	-	Classification of Cartridge Case Defects
* 9345013	-	Combination of Adopted Items (5,000 Cartridges, Cal .22: Long Rifle, Tracer, M861, Packed Level C)
9343015	-	Cartridge, Caliber .22 Long Rifle, Tracer, M861

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. The cartridge shall comply with all requirements specified on drawing (Dwg.) 9343015, all associated drawings and with all requirements specified in applicable specifications and standards.

* 3.3 Accuracy. The extreme spread measured from center to center of the extreme shots shall not exceed 45 millimeters. A group of 20 shots will be fired at a range of 22.86 meters.

3.4 Velocity. The mean velocity shall be 353 ± 23 m/sec at 6.0 meters from the muzzle. The standard deviation of the velocities from the mean velocity shall not exceed 12.2 m/sec.

3.5 Function and casualty. The malfunction rate shall not be greater than 4% of the rounds tested for function and casualty.

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3.6 Tracer evaluation. The tracer projectile must provide a visible trace during daylight and night conditions. The tracer projectile shall exhibit a visible trace of full luminosity from a point not greater than 8.2 meters from the muzzle end of the weapon to a point not less than 58.5 meters from the muzzle. 80% of the rounds shall trace.

3.7 Metal integrity. The cartridge shall be free of metal defects which include but are not limited to: folds, wrinkles,deep draw scratches, scaly metal, dents, perforations, and splits.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements for workmanship shall be as specified on applicable drawings, specifications and in accordance with the following:

3.9.1 Foreign matter. The cartridge shall be free of corrosion, stains, discoloration, dirt, oil, and smears of lacquer.

3.9.2 Cleaning. Cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent based upon examination with the unaided eye.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Classification of inspections. The following types of inspections shall be conducted as specified herein.

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspections to be performed. First article assemblies and components may be subjected by the Government to any or all of the examinations and tests specified in Table I and to any or all requirements of the applicable drawings.

4.3.3 Rejection. The First Article quantity shall be rejected upon failure of any assembly or component in the sample to comply with any visual or dimensional characteristic or upon failure to comply with the accept-reject criteria for any test. The accept-reject criteria for all tests shall be as specified in Tables I, II & IIA as applicable. The Government reserves the right to terminate its inspection whenever a component or assembly fails to comply with any visual or dimensional characteristic or components or assemblies fail to comply with the criteria for any test.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.4.2 Examination.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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b. Acceptable quality levels (AQL'S). AQL'S shall be as listed in the "Classification of Defects & Tests" paragraph.

c. Disposition. All non-conforming cartridges shall be classed defective and rejected. Non-conforming cartridge lots shall be rejected.

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TABLE I
FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63537A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*	Cartridge, Caliber .22, Long Rifle, Tracer, M861	1 OF 1	9343015
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects	20			
	Ballistic Tests				
	Velocity	30		3.4	
	Accuracy	60		3.3	
	Tracer Evaluation 1/			3.6	
	Other Firing Tests				
	Function and Casualty 1/				

NOTES:

1/ Sampling plans shall be in accordance with Table IIA.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63537A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
*	Cartridge, Caliber .22, Long Rifle Tracer, M861				9343015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
Major	1/				
101	Perforated case		2.5%	3.2	Visual
102	Case split		2.5%	3.2	
103	Case dent		2.5%	3.2	Visual
104	Cannelure missing or defective		2.5%	3.2	Visual
105	Discolored, dirty, oily, smeared		2.5%	3.2	Visual

NOTES:

1/ Dwg. 7643674 shall be used as a guide.

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE II
QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63537A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, Caliber .22, Long Rifle, Tracer, M861				9343015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ballistic Tests				
	Velocity	30		3.4	
	Accuracy	60		3.3	
	Tracer Evaluation 1/			3.6	
	Other Firing Tests				
	Function and Casualty 1/				

NOTES:

1/ Sampling plans shall be in accordance with Table IIA.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE IIA. Ballistic sample sizes.

Tracer Evaluation Test

First Sample			Cumulative (1st & 2nd Sample)		
Sample Size	Accept No.	Reject No.	Sample Size	Accept No.	Reject No.
240	40	50	480	70	71

Function and Casualty

Defect Description	First Sample			Cumulative (1st & 2nd Sample)		
	Sample Size	Accept No.	Reject No.	Sample Size	Accept No.	Reject No.
Bullet remaining in bore	240	0	1	480	0	1
Weapon stoppage	240	11	16	480	26	27

4.5 Methods of inspection.

4.5.1 Materials, components, and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 and Table I herein.

4.5.2 Velocity. The cartridges shall be fired from a M16 Rifle held in a Government approved test fixture. The velocity shall be measured over a range of 6 meters, the midpoint being at 6 meters from the muzzle. The velocity shall be measured with a 100 kilohertz (minimum) electronic counter chronograph using photoelectric screens spaced 6 meters apart with the first screen at 3 meters from the barrel.

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* 4.5.3 Accuracy. Three 20 shot groups shall be fired from a government approved test barrel fitted to a test action or in a selected match rifle of commercial manufacture held in a test fixture at a target 22.86 meters from the muzzle. The weapon is assembled and locked in position on the mount and boresighted into position. It is of prime importance that the assembly be mounted properly so that the weapon maintains its original position from shot to shot.

4.5.4 Tracer evaluation. Observation for trace shall be made at the weapon and at points 8.2 meters and 58.5 meters from the weapon. The test shall be conducted during daylight conditions. Three M16 Rifles shall be used with an equal number of rounds fired from each rifle. Lights or other suitable markers shall be placed opposite the 8.2 and 58.5 meter observation points to insure a line of sight from each point perpendicular to the plane of the trajectory. The observer, with line of sight established perpendicular to the trajectory by the light or visible marker shall note the trace performance of the bullet from each cartridge. The cartridge shall be considered defective if trace is not visible at 8.2 and 58.5 meters.

4.5.5 Function and casualty. Three M16 Rifles shall be used with an equal number of rounds fired from each rifle. The test shall be conducted at the same time as the Tracer test. All stoppages shall be recorded.

* 5. PACKAGING.

* 5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

* 5.1.1 Level A. Packaging for level A shall be in accordance with drawing 7553769.

* 5.1.2 Level B. Packaging for level B shall be the same as level A.

* 5.1.3 Level C. Packaging for level C shall be as specified on drawing 9345013.

* 5.2 Marking. Marking shall be in accordance with the applicable drawings specified in paragraph 5.1. When level C packaging is specified the boxes shall be marked with the ammunition lot number, national stock number, and other identification as shown in MIL-STD-129.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use in 5.56MM weapons.

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6.2 Ordering data. Ordering data should include:

a. Title, number and date of this specification.

b. Type and level of packing for the cartridges.

c. Provision for submission of Inspection Equipment Designs (DI-R-1714).

d. Provision for submission of detailed inspection plan for the cartridge (DI-R-4803) (see 6.7).

e. Provisions for the submission of acceptance inspection results for each lot of ammunition presented to the Government. (See 6.6)

* 6.3 Submission of inspection equipment designs for approval. Submit inspection equipment designs as required to Commander, ARDC, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801.

6.4 Hazard notice. The cartridge described herein and some of its components are flammable or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, and electrically indicted incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

* 6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is not under the cognizance of ARDC.

* 6.6 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US Army Armament Research and Development Center (ARDC), one (1) copy of all ballistic data and the ammunition data card for each lot shall be forwarded to: Commander, ARDC, ATTN: DRSMC-QAF-S(D), Dover, NJ 07801.

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* 6.7 Submission of detailed inspection plan for review. The detailed inspection plan for the cartridge shall be submitted to Commander, ARDC, ATTN: DRSMC-QAF(D), Dover, NJ 07801. Submission shall be 30 days after specification is made contractual. All changes to the plan shall be submitted to the above address for review prior implementation.

* 6.8 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-63537A	CARTRIDGE, CALIBER .22 LONG RIFLE, TRACER, M861

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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